Credit Risk: Part II

Lecture 6

BANK3011

DR WEI CUI

THE UNIVERSITY OF SYDNEY
Business School



Objectives of this week's lecture

- We identify the issues involved in managing loan portfolios
  - ✓ Measuring the level of loan concentration risk
  - ✓ Application of Modern Portfolio Theory (MPT)
  - ✓ Diversification effect on loan portfolios

- The use of portfolio models in the management of credit risk
  - ✓ Credit rating migration and portfolio correlations
  - ✓ Moody's KMV model
  - ✓ Other methods based on partial application of MPT

1. Loan concentration risk

- Concentration risk in general: Concentration risk arises when a bank has large exposures to a particular borrower, industry, geographic region, asset class, or funding source, making it vulnerable to losses from a common adverse shock.

Case	Main concentration	Shock	Main transmission
Penn Square (1982)	Oil & gas **loans**	Energy-sector downturn	Borrower defaults → credit losses → failure
SVB (2023)	Tech/VC **depositors** + long-duration securities	Rising rates + deposit run	Securities losses/liquidity pressure → failure

1. Loan concentration risk

- Migration Analysis:
  - ✓ The allocation of obligors to rating grades is useful in identifying and reducing loan portfolio concentration
  - ✓ Ratings of loan pools or certain sectors are tracked.
    - ➢ If actual rating deteriorates faster than historical experience, curtail lending to that sector or loan class.
  - ✓ Historical credit migration measured through loan migration or transition matrices.
    - ➢ Identify how creditworthiness might change in the future

1. Loan concentration risk

- Migration Analysis:

TABLE 9.1 A hypothetical rating migration or transition matrix

		Risk grade at end of year			
		1	2	3	D[a]
Risk grade at beginning of year	1	0.85	0.10	0.04	0.01
	2	0.12	0.83	0.03	0.02
	3	0.03	0.13	0.80	0.04

[a] D = default.

1. Loan concentration risk

- Moodys - Transition Matrix

Figure 11.14 Average One-Year Rating Transition Matrix, 1920–1996

		Rating To: Aaa	Aa	A	Baa	Ba	B	Caa-C	Default	WR
Rating From:	Aaa	88.32%	6.15%	0.99%	0.23%	0.02%	0.00%	0.00%	0.00%	4.29%
	Aa	1.21%	86.76%	5.76%	0.66%	0.16%	0.02%	0.00%	0.06%	5.36%
	A	0.07%	2.30%	86.09%	4.67%	0.63%	0.10%	0.02%	0.12%	5.99%
	Baa	0.03%	0.24%	3.87%	82.52%	4.68%	0.61%	0.06%	0.28%	7.71%
	Ba	0.01%	0.08%	0.39%	4.61%	79.03%	4.96%	0.41%	1.11%	9.39%
	B	0.00%	0.04%	0.13%	0.60%	5.79%	76.33%	3.08%	3.49%	10.53%
	Caa-C	0.00%	0.02%	0.04%	0.34%	1.26%	5.29%	71.87%	12.41%	8.78%

1. Loan concentration risk

- Migration Analysis:
  - ✓ Step 1: sorting loans into different groups based on credit risk (measured by external or internal rating) levels
  - ✓ Step 2: calculating the proportions of loans in each (beginning-of-year) group at various credit risk levels at the end of year.
  - ✓ Step 3: comparing the actual credit deterioration with the historical numbers. If the credit risk of a sector deteriorates faster than historical experience, then curtail lending to that sector or loan class.
- Information on migration analysis:
  - ✓ Standard & Poor's: www.standardandpoors.com
  - ✓ Moody's KMV: www.moodyskmv.com

1. Loan concentration risk

- Concentration Limits:
  - ✓ External limits set on the maximum loan size to individual borrowers/sectors/segments
    - ➢ Eg. Max loan to individuals = 10% of capital
  - ✓ Used to reduce exposures to certain industries and to increase exposure to others.
  - ✓ Use of aggregate limits for industries in which performance is highly correlated.
  - ✓ Additional use of geographical limits.

1. Loan concentration risk

- Calculating Concentration Limits for a Loan Portfolio:

$$\text{Concentration limit} = \text{Maximum loss as a percentage of capital} \times \frac{1}{\text{Loss rate}}$$

Note: Maximum loss expressed as percent of capital

1. Loan concentration risk

- Using concentration limits - An example
  - ✓ An FI has set a maximum loss of 2 percent of total capital as a basis for setting concentration limits on loans to individual firms. If it has set a concentration limit of 25 percent to a firm, what is the expected loss rate for that firm?

  - ✓ Concentration limit = Maximum loss as a percent of capital / Loss rate

    25 percent = 2 percent / Loss rate => Loss rate = 0.02/0.25 = 8 percent

  - ✓ The expected loss rate for the firm is 8 percent.

1. Loan concentration risk

- Concentration limits - sample question
  - ✓ If a bank's concentration limit (as a percent of capital) is 20 percent, and its expected recovery from defaulted loans is 70 percent, what is the maximum loss it permits to affect its capital in the event of a default?

1. Loan concentration risk

- Concentration limits - sample question
  - ✓ If a bank's concentration limit (as a percent of capital) is 20 percent, and its expected recovery from defaulted loans is 70 percent, what is the maximum loss it permits to affect its capital in the event of a default?

  - ➢ Concentration limit = Maximum loss as a percent of capital × (1 ÷ Loss rate)
  - ➢ Maximum loss as a percent of capital = Concentration limit × Loss rate
  - ➢ Loss rate = (1 – recovery rate)
  - ➢ ML = 0.20 × 0.30 = 0.06

2. Modern Portfolio Theory (MPT)

- Expected return (R_p) on a portfolio of assets:

$$\bar{R}_p = \sum_{i=1}^{n} X_i \ \bar{R}_i$$

Where:
R_i = expected returns of each bond or loan, and
X_i = the proportion of the asset portfolio invested in the ith asset.

- Variance of returns or risk of the portfolio (σ^2_p):

$$\sigma_p^2 = \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{X}_\mathrm{i}^2 \sigma_i^2 + \sum_{i=1}^{n} \sum_{\substack{i=1 \\ i \neq j}}^{n} \mathrm{X}_\mathrm{i} X_j \rho_{ij} \sigma_i \sigma_j$$

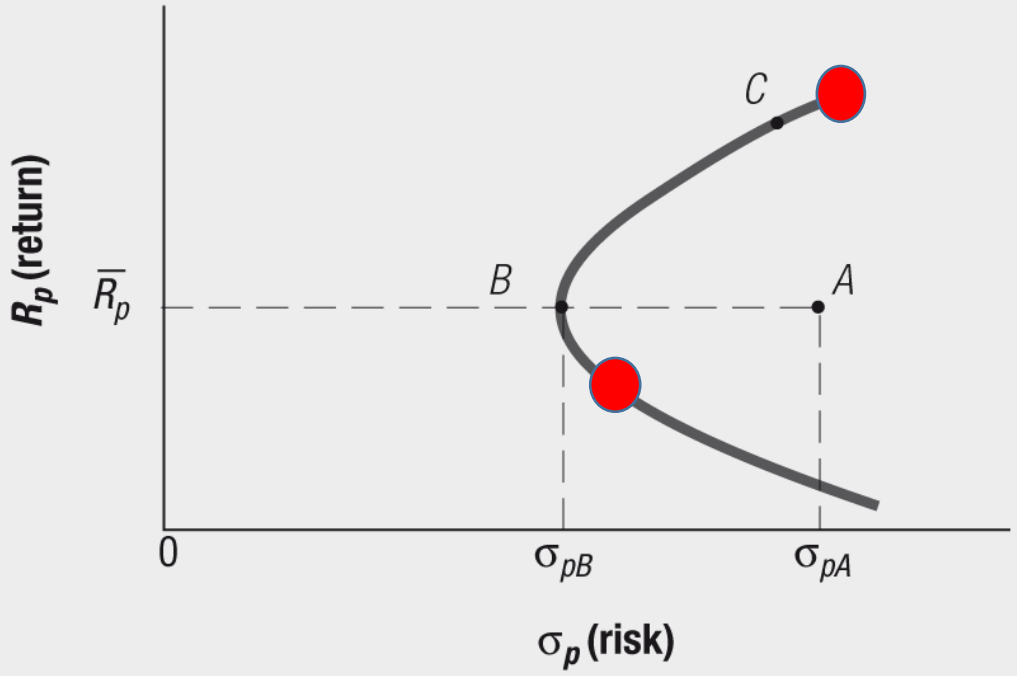
Where:
σ^2_i = the variance of returns on the ith asset,
ρ_{ij} = the correlation between the returns on the ith and jth asset,
$\rho_{ij}\ \sigma_i\ \sigma_j$ = the covariance of returns between the ith and jth asset,

2. Modern Portfolio Theory (MPT)

- Fundamental lesson of MPT:
  - ✓ Diversification benefit if there is imperfect correlation of the returns of different assets (loans for FIs!)
- Minimum Risk Portfolio:
  - ✓ Combination of assets that reduces the variance of portfolio returns on the lowest feasible level given a specific expected return.

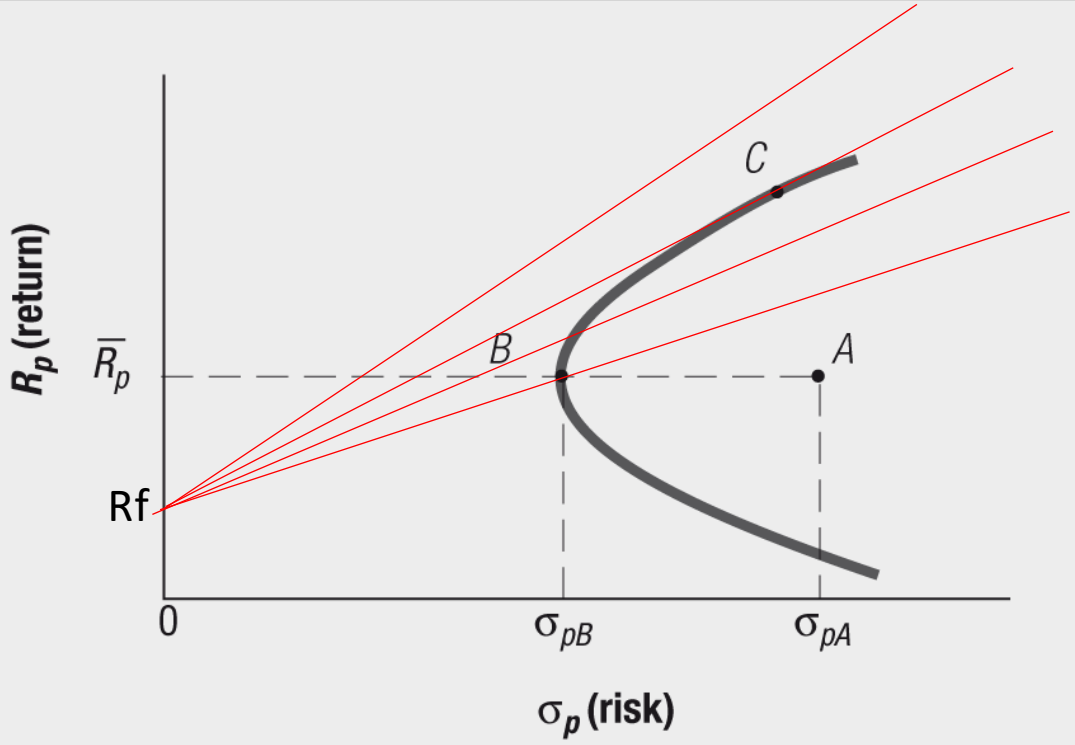
- Minimum Risk Portfolio:
  - ✓ Combination of assets that reduces the variance of portfolio returns on the lowest feasible level given a specific expected return.

FIGURE 9.1 FI portfolio diversification



- Minimum Risk Portfolio:
  - ✓ Combination of assets that reduces the variance of portfolio returns on the lowest feasible level given a specific expected return.

FIGURE 9.1 FI portfolio diversification



2. Modern Portfolio Theory (MPT)

- MPT: Application
  - ✓ Optimization of loan portfolios
    - ➢ We can basically view a bank issuing loans as an investor
    - ➢ The objective is to choose an optimal allocation (weights) for all possible loans
    - ➢ maximize the return of the loan portfolio, while at the same time minimize the risk (volatility of return) of the loan portfolio.
  - ✓ In terms of credit risk management
    - ➢ The credit risk could be reduced via diversification if there is no perfect correlation between default risks of individual assets.

3. Loan portfolios

- Suppose that Bank of USYD holds two loans with the following characteristics:

Loan i	X_i	R_i	σ_i	σ_i^2
1	0.55	8%	8.55%	0.731%
2	0.45	10 %	9.15 %	0.8372%

$\rho_{12} = 0.24$

- and Xi is the proportion of the bank's loan portfolio in each loan i

Calculate the return and risk of the FI's loan portfolio…..

- And the answers are…
  - ✓ The expected return on the loan portfolio is:

$$\bar{R}_p = \sum_{i=1}^{n} X_i \bar{R}_i = 0.55 * 0.08 + 0.45 * 0.1 = 0.089$$

  - ✓ The risk of the loan portfolio is:

$$\sigma_p^2 = \sum_{i=1}^{n} X_i^2 \sigma_i^2 + \sum_{i=1}^{n} \sum_{\substack{i=1 \\ i \neq j}}^{n} X_i X_j \rho_{ij} \sigma_i \sigma_j = 0.004836$$

$$\sigma_p = 0.069542$$

4. KMV Portfolio Manager Model

- Developed by KMV Corporation (now Moody's KMV)
- Applies MPT to the loan portfolio, although many loans have non-traded aspects
- Required input variables:
  - ✓ Expected return on a loan to borrower i (R_i),
  - ✓ Risk of a loan to borrower i (σ_i), and
  - ✓ Correlation of default risks between loan made to borrowers i and j (ρ_{ij}).

4. KMV Portfolio Manager Model

- Return on a loan to borrower i (Ri)
  - ✓ AIS, if the client does not default, which occurs at prob (1-EDF)
  - ✓ AIS – LGD, if the client defaults, which occurs at prob EDF

where

- ✓ AIS is the all-in-spread, annual fees + the difference between loan rate paid by the borrower minus the FI's funding cost
- ✓ LGD is the loss if the client defaults (Loss Given Default), calculated as the contractually promised return multiplying by the (1 – recovery rate)
- ✓ EDF is the probability of default, termed as expected default frequency by KMV

4. KMV Portfolio Manager Model

- Expected (excess) return on a loan to borrower I

$$R_i = AIS_i \times (1 - EDF_i) + (AIS_i - LGD_i) \times EDF_i$$

$$= AIS_i - EDF_i \times LGD_i$$

where ($EDF_i \times LGD_i$) is the expected loss (EL_i).

- Risk of a loan to borrower i (σi)

$$\sigma_i = \sqrt{\mathrm{EDF_i}\left(1 - EDF_i\right)} \times \mathrm{LGD_i}$$

- Correlation between returns of loans made to borrowers i and j: estimated as correlation between the common return components of the asset returns of borrowers i and j

  ✓ Extent of correlation is related to industry, geography and time

4. KMV Portfolio Manager Model

- An application of KMV
  - ✓ SydneySide Bank uses the Moody's Analytics Portfolio Manager model to evaluate the risk-return characteristics of the loans in its portfolio. A specific $10 million loan earns 2 percent per year in fees and the loan is priced at a 4 percent spread over the cost of funds for the bank. Because of collateral considerations, the loss to the bank if the borrower defaults will be 20 percent of the loan's face value. The expected probability of default is 3 percent.
  - ✓ What is the anticipated return on this loan?
  - ✓ What is the risk of the loan?

4. KMV Portfolio Manager Model

- An application of KMV

Expected return = $AIS_i – E(L_i) = (0.02 + 0.04) – (0.03 \times 0.20) = 0.054$ or 5.4 percent

Risk of the loan = $\sigma_{Di} \times LGD_i = [0.03 \times (0.97)]^{½} \times 0.20 = 0.0341$ or 3.41 percent

4. KMV Portfolio Manager Model

- An application of KMV – sample question
  - ✓ Suppose that a FI holds two loans with the following characteristics (from Saunders and Cornett, Ch12)

Loan i	X_i	Annual Spread between Loan Rate and FI's Cost of Funds	Annual Fees	Loss to FI Given Default	Expected Default Frequency	
1	.60	5.0%	2.0%	25%	3%	$\rho_{12} = -.25$
2	.40	4.5	1.5	20	2	

  - ✓ Calculate the return and risk of the portfolio

4. KMV Portfolio Manager Model

- An application of KMV – sample question
  - ✓ Calculate the return and risk of the portfolio

The return and risk on loan 1 are:

$$R_1 = (.05 + .02) - [.03 \times .25] = 0.0625 \text{ or } 6.25\%$$
$$\sigma_1 = [\sqrt{.03(.97)}] \times .25 = .04265 \text{ or } 4.265\%$$

The return and risk on loan 2 are:

$$R_2 = (.045 + .015) - [.02 \times .20] = 0.056 \text{ or } 5.60\%$$
$$\sigma_2 = [\sqrt{.02(.98)}] \times .20 = .028 \text{ or } 2.80\%$$

4. KMV Portfolio Manager Model

- An application of KMV – sample question
  - ✓ Calculate the return and risk of the portfolio

The return and risk of the portfolio are then:

$$R_p = .6(6.25\%) + .4(5.60\%) = 5.99\%$$

$$\sigma_p^2 = (.6)^2(.04265)^2 + (.4)^2(.028)^2 + 2(.6)(.4)(-.25)(.04265)(.028) = .0006369$$

thus, $\sigma_p = \sqrt{.0006369} = .0252 = 2.52\%.$

5. Loan volume

- Partial Applications of MPT: Loan-volume
  - ✓ Comparing the asset allocation in the loan portfolios to the market benchmarks.
  - ✓ Deviations from the market portfolio benchmark indicate the relative degree of loan concentration.

5. Loan volume

- Partial Applications of MPT: Loan-volume

$$\sigma_j = \sqrt{\frac{\sum_{i=1}^{N}\left(X_{ij} - X_i\right)^2}{N}}$$

σ_j = standard deviation of bank's *j*th asset allocation proportions from the national benchmark,

X_{ij} = asset allocation proportions of the *j*th bank,

X_i = national asset allocations,

N = number of observations or loan categories.

5. Loan volume

- Loan volume based model (cont)
  - ✓ Benchmark Data:
    - ➢ Commercial bank call reports
      - Can be aggregated to estimate national allocations
    - ➢ Data on shared national credits
      - National database that breaks commercial and industrial loan volume into 2-digit SIC codes
    - ➢ Commercial databases

- An example using loan volume

Assume that, on average, national banks engaged primarily in mortgage lending have their assets diversified in the following proportions: 60 percent residential, 15 percent commercial, 5 percent international, and 20 percent mortgage-backed securities. A local bank has the following distribution of mortgage loans: 50 percent residential, 30 percent commercial, and 20 percent international. How does the local bank differ from national banks?

Using Xs to represent portfolio holdings:

$$(X_{1j} - X_1)^2 \qquad (0.50 - 0.60)^2 = 0.0100$$
$$(X_{2j} - X_2)^2 \qquad (0.30 - 0.15)^2 = 0.0225$$
$$(X_{3j} - X_3)^2 \qquad (0.20 - 0.05)^2 = 0.0225$$
$$(X_{4j} - X_4)^2 \qquad \underline{(0.00 - 0.20)^2 = 0.0400}$$

$$\sum_{i=1}^{n} (X_{ij} - X_i)^2 \qquad \sum_{i=1}^{n=4} = 0.0950$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (X_{ij} - X_i)^2}{n}} \qquad \sigma = 15.41 \text{ percent}$$

The bank's standard deviation in its loan portfolio allocation is 15.41 percent. This suggests that the bank is different from the national average. Whether it is significantly different cannot be stated without comparing it to other banks.

5. Loan volume

- Loan volume based model – sample question (example 12-4 from the text book)

	(1) National	(2) Bank A	(3) Bank B
Real estate	45%	65%	10%
C&I	30	20	25
Individuals	15	10	55
Others	10	5	10
	100%	100%	100%

- 1. Calculate deviation in bank A's portfolio allocation using national average as benchmark
- 2. Calculate deviation in bank B's portfolio allocation using national average as benchmark
- 3. Comparing bank A and B, which bank deviates more from the benchmark?

5. Loan volume

- Loan volume based model – sample question (example 12-4 from the text book)

  - Deviation in bank A's portfolio

$$(X_{1A} - X_1)^2 = (.65 - .45)^2 = .0400$$
$$(X_{2A} - X_2)^2 = (.20 - .30)^2 = .0100$$
$$(X_{3A} - X_3)^2 = (.10 - .15)^2 = .0025$$
$$(X_{4A} - X_4)^2 = (.05 - .10)^2 = .0025$$

and

$$\sum_{i=1}^{4} = .0550$$

Therefore, $\sigma_A = (.0550/4)^{1/2} = 11.73\%$.

5. Loan volume

- Loan volume based model – sample question (example 12-4 from the text book)

  - Deviation in bank B's portfolio

$$(X_{1B} - X_1)^2 = (.10 - .45)^2 = .1225$$
$$(X_{2B} - X_2)^2 = (.25 - .30)^2 = .0025$$
$$(X_{3B} - X_3)^2 = (.55 - .15)^2 = .1600$$
$$(X_{4B} - X_4)^2 = (.10 - .10)^2 = .0000$$

and

$$\sum_{i=1}^{4} = .2850$$

Therefore, $\sigma_B = (.2850/4)^{1/2} = 26.69\%$. As you can see, bank B deviates more significantly from the national benchmark than bank A because of its heavy concentration on loans to individuals.

5. Loan volume

- Partial Applications of MPT: Loan loss ratio
  - ✓ Based on the implication of CAPM that an asset's risk is determined by its systematic risk – how its return co-move with market return
  - ✓ Estimates systematic loan loss risk by industry sector relative to the total loan portfolio.
  - ✓ Use of time-series regression of quarterly losses:

$$\frac{\text{Sectoral losses in the ith sector}}{\text{Loans to the ith sector}} = \alpha + \beta_i \left(\frac{\text{Total loan losses}}{\text{Total loans}} \right)$$

α = loan loss rate for a sector with no sensitivity to losses on the aggregate portfolio,

β_i = systematic loss sensitivity of the *i*th sector loans to total loans.

6. Regulatory models

- Historically, concentration limits applied, but were criticised for being arbitrary.

- Introduction of Basel II since mid 2000s has led to a greater reliance on quantitative models, such as:
  - ✓ Moody's KMV
  - ✓ CreditMetrics (JPM) (Riskmetrics owned by Reuters) – Mark-to-market paradigm: estimate VaR on loans and loan portfolio
  - ✓ Credit Risk+ (CSFP) - Default paradigm: estimate the expected loss of loans and the distribution of those losses to understand how much capital is required to meet those losses

6. Regulatory models

- Regulators Views on Credit Models – Bank for International Settlements
  - ✓ Provide timely data
  - ✓ Encourage the centralisation of data
  - ✓ Are effective in reflecting risk concentration. Provided correlations are well understood VaR can provide an aggregate value
    - ➢ History shows that the calculation of credit correlations is extremely difficult and are not well understood, e.g. Sub-prime
  - ✓ Possibly a more responsive informational tool

6. Regulatory models

- Regulators Views on Credit Models – Bank for International Settlements
  - ✓ Foster enhanced data collection
  - ✓ Facilitate more informed limit setting
  - ✓ Assist in the better pricing of risk
  - ✓ Provide a formal framework for the allocation of economic capital

Summary

- Loan portfolios present additional credit risk concerns for FIs
  - ✓ Concentration risk
- There are various methods to measure the level of loan concentration
- Useful to apply MPT to manage loan portfolios
- Credit portfolio management is critical for managing the extent of portfolio concentration risks